Data Sciences – CentraleSupelec
Large Scale and Distributed Optimization
Part I: Introduction

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Optimization ?

Whatever people do, at some point they get a craving to organize things in a best possible way. This intention, converted in a mathematical form, turns out to be an optimization problem of certain type.
(Yurii Nesterov)



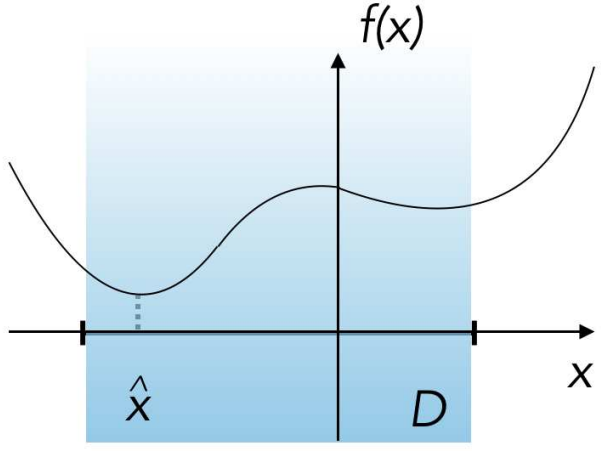
Optimization problems

- Minimization problems
  $f : D \mapsto [-\infty, +\infty]$, $D \subset \mathbb{R}^N$: cost function
  We want to

$$\begin{aligned} & \text{Find } \widehat{x} \in D \quad \text{such that} \quad (\forall x \in D)\ f(\widehat{x}) \leq f(x) \\ \Leftrightarrow \quad & \text{Find } \widehat{x} \in D \quad \text{such that} \quad f(\widehat{x}) = \inf_{x \in D} f(x) \end{aligned}$$

that is

$$\text{Find } \widehat{x} \in \operatorname*{Argmin}_{x \in D} f(x).$$



Optimization problems

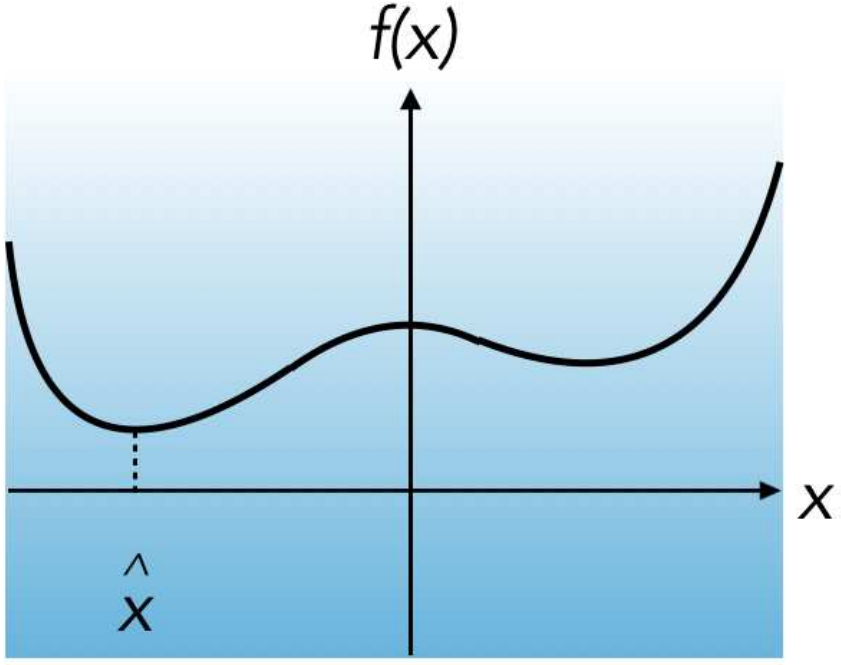
- Maximization problems
  f: reward function
  We want to

$$\begin{aligned} & \text{Find } \widehat{x} \in D \quad \text{such that} \quad (\forall x \in D)\ f(\widehat{x}) \geq f(x) \\ \Leftrightarrow \quad & \text{Find } \widehat{x} \in D \quad \text{such that} \quad (\forall x \in D)\ -f(\widehat{x}) \leq -f(x) \\ \Leftrightarrow \quad & \text{Find } \widehat{x} \in \operatorname*{Argmin}_{x \in D} \big(-f(x)\big). \end{aligned}$$

Without loss of generality, we can focus on minimization problems with $f : D \to]-\infty, +\infty]$.

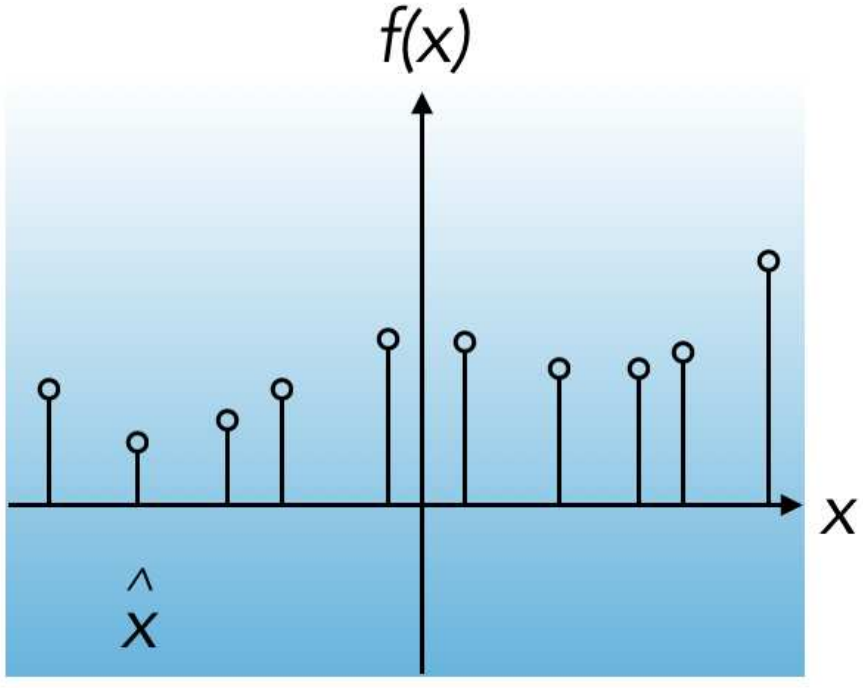
Various types of minimization problems

- $D = \mathbb{R}^N$: unconstrained problem



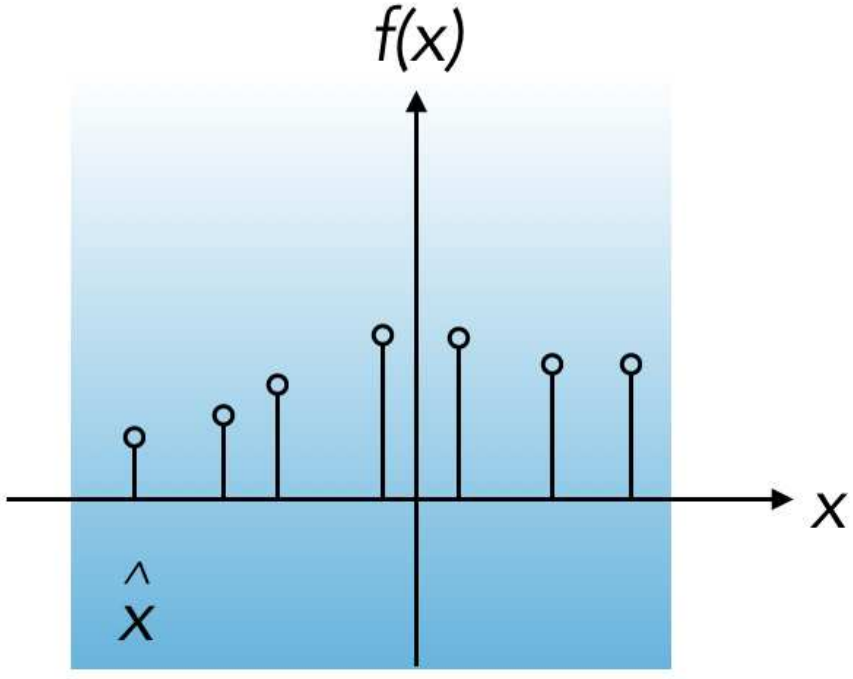
Various types of minimization problems

- $D = \mathbb{R}^N$: unconstrained problem
- D countable: discrete optimization problem



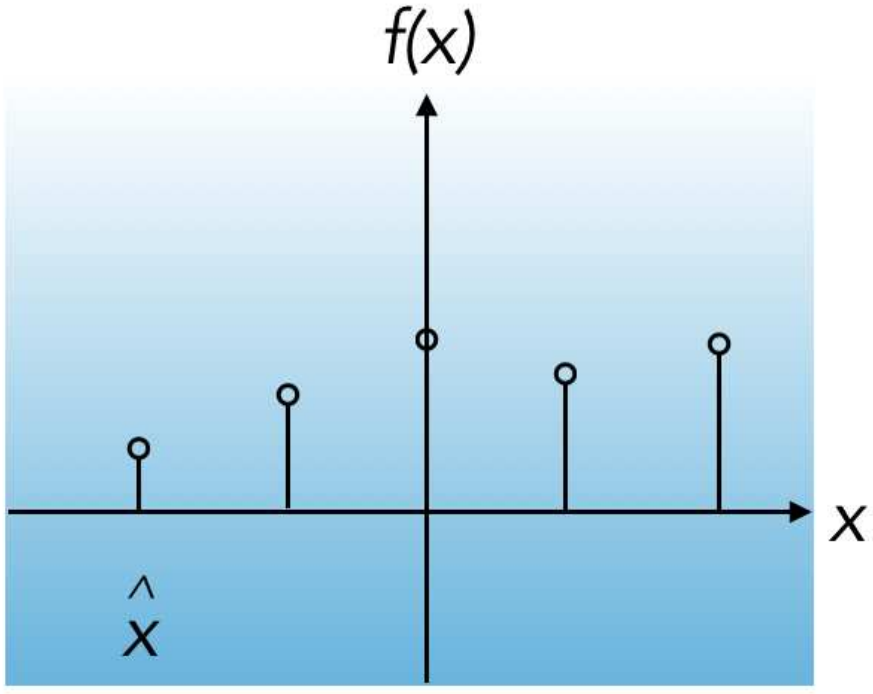
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- $D = \mathbb{R}^N$: unconstrained problem
- D countable: discrete optimization problem
  - D finite: combinatorial optimization problem



Various types of minimization problems

- $D = \mathbb{R}^N$: unconstrained problem
- D countable: discrete optimization problem
  - D finite: combinatorial optimization problem
  - $D \subset \mathbb{Z}^N$: integer optimization problem



Various types of minimization problems

- D uncountable: continuous optimization problem
- Optimization problem with P equality constraints and Q inequality constraints:

$$D = \{x \in \mathbb{R}^N \mid (\forall i \in \{1, \dots, P\})\ \varphi_i(x) = \delta_i \\ \text{and } (\forall j \in \{1, \dots, Q\})\ \psi_j(x) \leq \eta_j\}$$

where $(\forall i \in \{1, \dots, P\})\ \delta_i \in \mathbb{R}$ and $\varphi_i \colon \mathbb{R}^N \to]-\infty, +\infty]$,
$(\forall j \in \{1, \dots, Q\})\ \eta_j \in \mathbb{R}$ and $\psi_j \colon \mathbb{R}^N \to]-\infty, +\infty]$.
If $\varphi_i \colon x \mapsto \langle x \mid u_i \rangle$ with $i \in \{1, \dots, P\}$ and $u_i \in \mathbb{R}^N$, then *linear* (or affine) equality constraint.
If $\psi_j \colon x \mapsto \langle x \mid u_j \rangle$ with $j \in \{1, \dots, Q\}$ and $u_j \in \mathbb{R}^N$, then *linear* (or affine) inequality constraint.
If, in addition, f is linear, then linear programming problem.

Various types of minimization problems

Remark:

$$\begin{aligned} & \text{Find } \widehat{x} \in \operatorname*{Argmin}_{x \in D} f(x) \\ \Leftrightarrow \quad & \text{Find } \widehat{x} \in \operatorname*{Argmin}_{x \in \mathbb{R}^N} \widetilde{f}(x) \end{aligned}$$

where

$$(\forall x \in \mathbb{R}^N) \quad \widetilde{f}(x) = \begin{cases} f(x) & \text{if } x \in D \\ +\infty & \text{otherwise.} \end{cases}$$

Allowing non finite valued functions leads to a unifying view of constrained and unconstrained minimization problems.

Main questions to be addressed

1. Existence/uniqueness of a solution $\widehat{x}$?

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3. Designing an algorithm to approximate a solution in the frequent case when no closed form solution is available, i.e. building a sequence $(x_n)_{n\in\mathbb{N}}$ of $\mathbb{R}^N$ such that

$$\lim_{n\to+\infty} x_n = \widehat{x}.$$

Main questions to be addressed

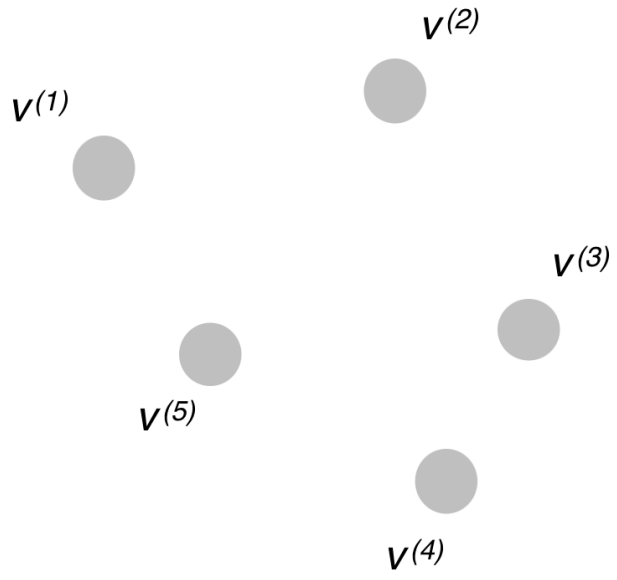
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4. Evaluation of the performance of the optimization algorithm:
   - Convergence speed
     Example: If there exists $\rho \in]0,1[$ and $n^* \in \mathbb{N}$ such that $(\forall n \geq n^*)$ $\|x_{n+1} - \widehat{x}\| \leq \rho \|x_n - \widehat{x}\|$, then *Q-linear* convergence rate.
     If $\lim_{n\to+\infty} \frac{\|x_{n+1}-\widehat{x}\|}{\|x_n-\widehat{x}\|} = 0$, then *Q-superlinear* convergence rate.
   - Robustness to numerical errors
   - Amenability to parallel/distributed implementations.

Valued graphs

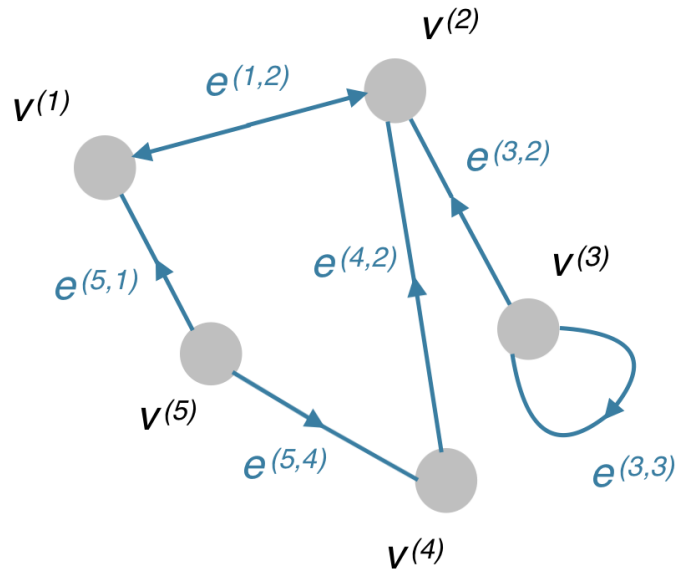
- $\mathcal{V} = \left\{ v^{(i)} \;\middle|\; i \in \{1, \ldots, M\} \right\}$
  $\rightsquigarrow$ set of vertices $=$ objects
  $v^{(i)} \in \mathcal{V} \leftrightarrow i \in \{1, \ldots, M\}$



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- $\mathcal{E} = \left\{ e^{(i,j)} \;\middle|\; (i,j) \in \mathbb{E} \right\}$
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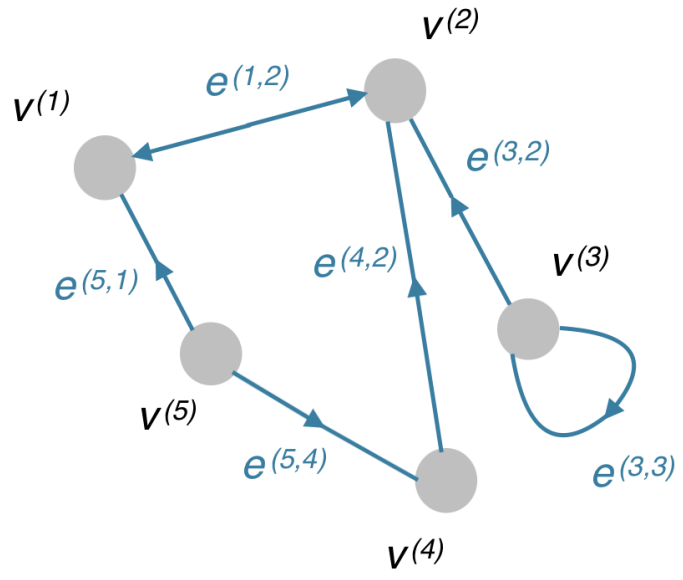


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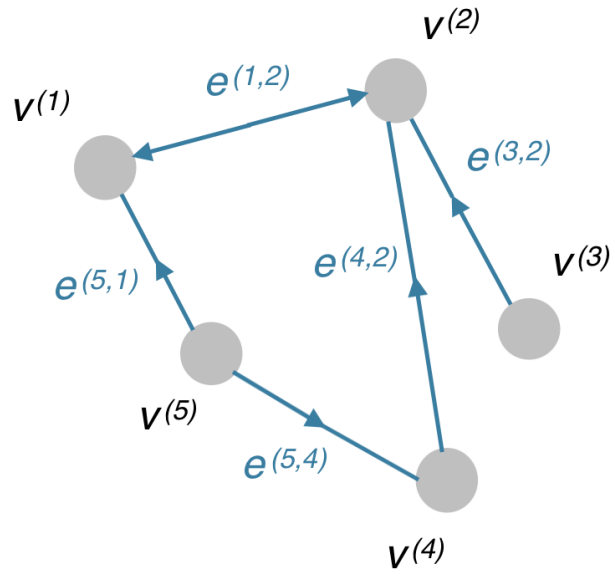
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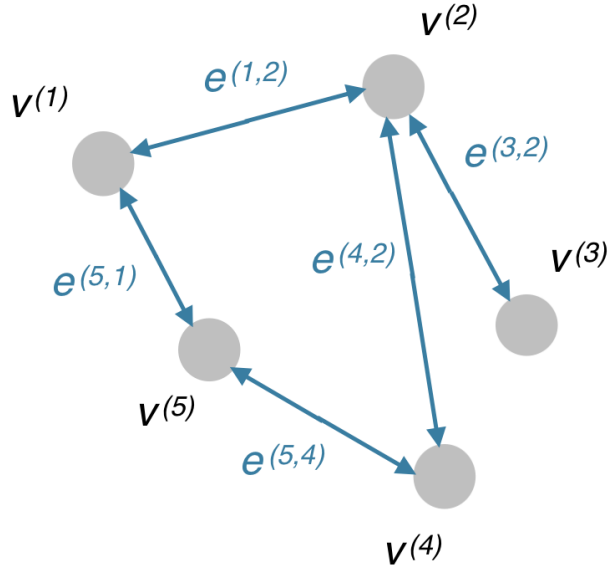


- directed reflexive graph: $|\mathbb{E}| \leq M^2$
  directed nonreflexive graph: $|\mathbb{E}| \leq M(M-1)$

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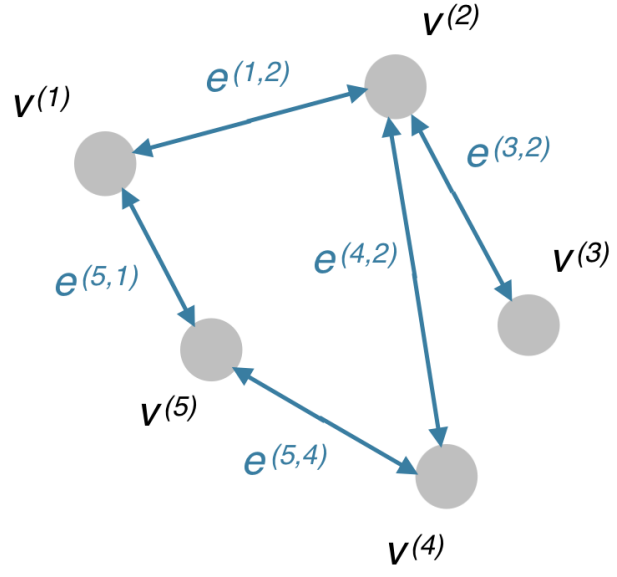
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- directed reflexive graph: $|\mathbb{E}| \leq M^2$
  directed nonreflexive graph: $|\mathbb{E}| \leq M(M-1)$
  undirected nonreflexive graph: $|\mathbb{E}| \leq M(M-1)/2$

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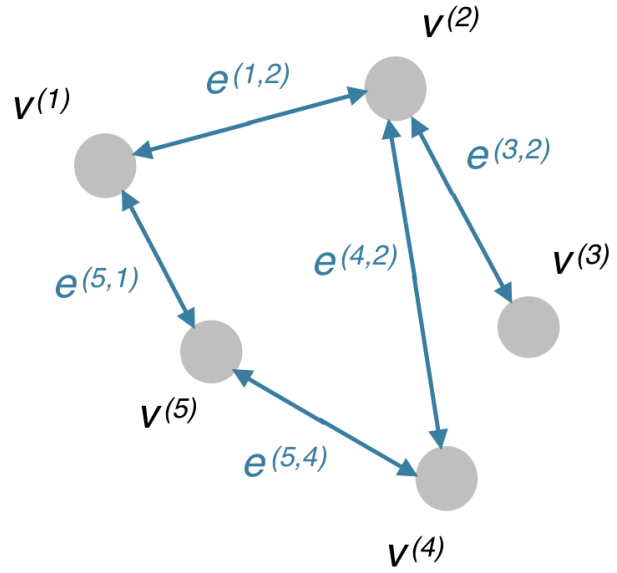


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- directed reflexive graph: $|\mathbb{E}| \leq M^2$
  directed nonreflexive graph: $|\mathbb{E}| \leq M(M-1)$
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- $(x^{(i)})_{1 \leq i \leq M}$: weights on vertices (scalars or vectors)

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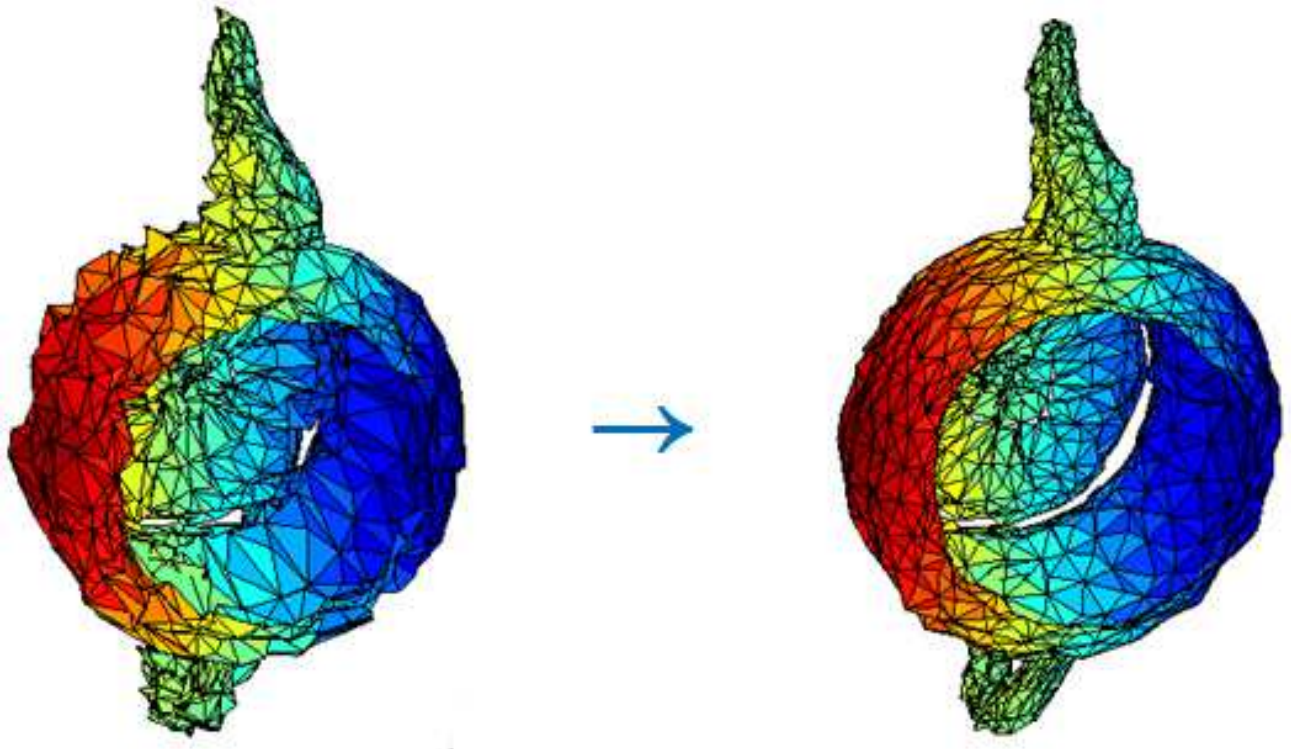


- $\mathcal{E} = \left\{ e^{(i,j)} \;\middle|\; (i,j) \in \mathbb{E} \right\}$
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- $(x^{(i)})_{1 \leq i \leq M}$: **weights on vertices** (scalars or vectors)
- $(x^{(i,j)})_{(i,j) \in \mathbb{E}}$: **weights on edges** (scalars or vectors)

Example: Mesh denoising problem

- Remove uncertainties in mesh measurements



Example: Mesh denoising problem

- Undirected nonreflexive graph

  $\mathcal{V}$: set of vertices of the mesh

  $\mathcal{E}$: set of edges of the mesh

  $x = (x^{(i)})_{1 \leq i \leq M}$ where, for every $i \in \{1, \ldots, M\}$,

  $x^{(i)} \in \mathbb{R}^3$: 3D coordinates of the i-th vertex of the true object

  $N = 3M$ and $D = \mathbb{R}^N$.

- Cost function

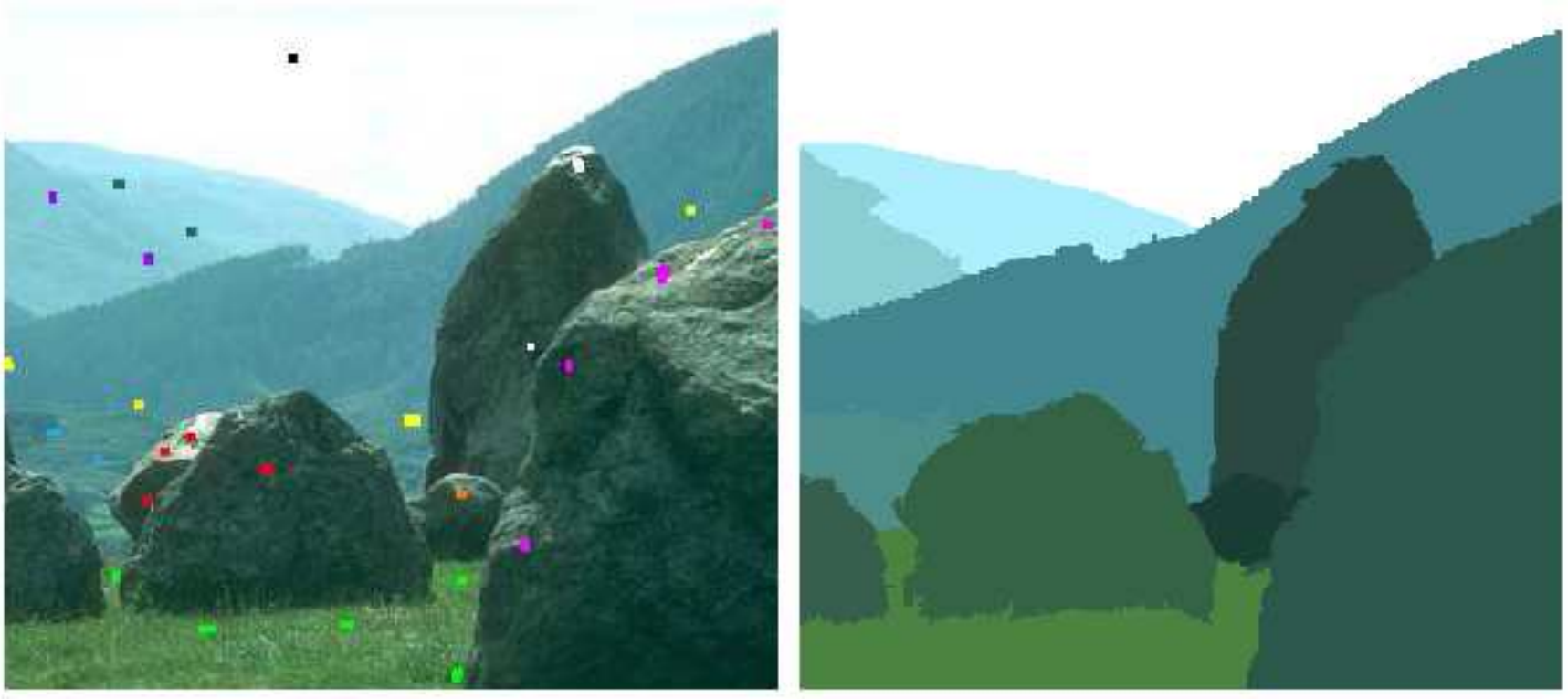
$$f(x) = \sum_{i=1}^{M} \left(\|x^{(i)} - y^{(i)}\|^2 + \Big(\sum_{j \in \mathcal{N}_i} \omega_{i,j} \|x^{(i)} - x^{(j)}\|^2 \Big)^{1/2} \right)$$

  where, for every $i \in \{1, \ldots, M\}$,

  - $y^{(i)}$: 3D measured coordinates of the i-th vertex
  - $\mathcal{N}_i$: neighborhood of i-th vertex
  - $(\omega_{i,j})_{j \in \mathcal{N}_i}$: nonnegative regularization constants.

Example: Segmentation problem

- Assign a label to every pixel in an image so that pixels with the same label share certain characteristics.



Example: Segmentation problem

- Undirected nonreflexive regular graph

  $\mathcal{V}$: Set of pixels

  $\mathcal{E}$: Set of neighborhood relationships between pixels

  $x = (x^{(i)})_{1 \leq i \leq M}$ where, for every $i \in \{1, \ldots, M\}$, $x^{(i)} \in \mathcal{L}$: label of pixel i

  $\mathcal{L} \subset \mathbb{R}$: finite set of labels

  $N = M$ and $D = \mathcal{L}^M$.

- Cost function

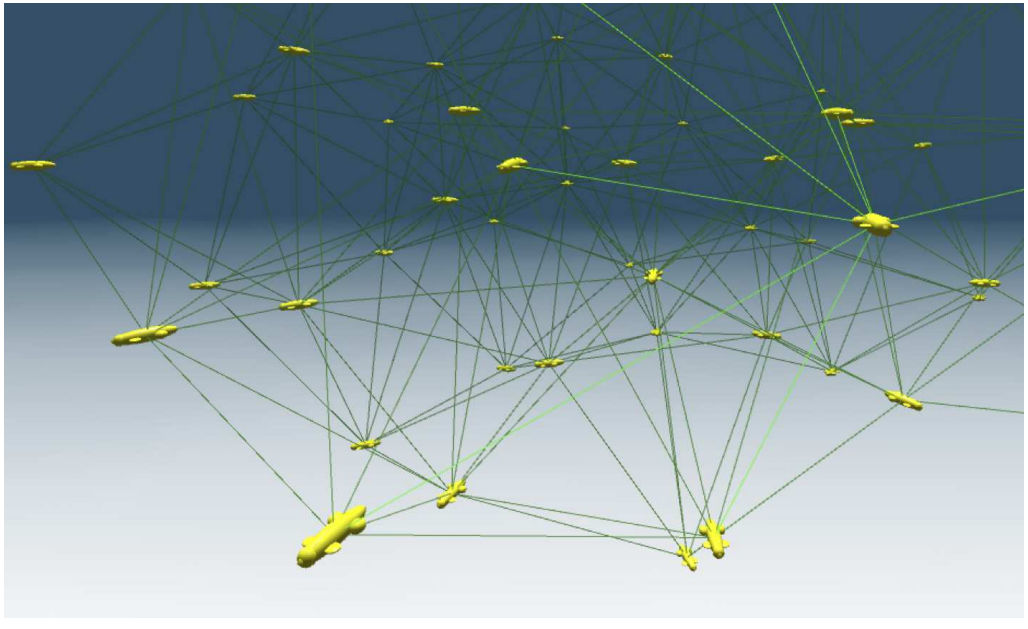
$$f(x) = \sum_{i=1}^{M} \alpha_i^p |x^{(i)} - y^{(i)}|^p + \sum_{(i,j) \in \mathbb{E}} \omega_{i,j}^q |x^{(i)} - x^{(j)}|^q$$

  where, for every $i \in \{1, \ldots, M\}$,
  - $y^{(i)}$: intensity value of pixel i
  - $(\alpha_i)_{1 \leq i \leq M}$ and $(\omega_{i,j})_{(i,j) \in \mathbb{E}}$: nonnegative constants
  - $(p, q) \in [1, +\infty[^2$.

Example: Multiagent consensus problem

- Each agent processes local data and cooperates with its neighbors to estimate global parameters.

  Examples: Trajectory planning, distributed clustering, Internet traffic anomalies, ...



Example: Multiagent consensus problem

- Directed nonreflexive graph

  $\mathcal{V}$: Set of agents

  $\mathcal{E}$: Set of relationships between agents

  $x = (x^{(i)})_{1 \leq i \leq M}$ vector of estimates in $\mathbb{R}^P$ at each node

  $N = MP$ and

$$D = \left\{ (x^{(i)})_{1 \leq i \leq M} \in (\mathbb{R}^P)^M \;\middle|\; x^{(1)} = \ldots = x^{(M)} \right\}.$$

- Cost function

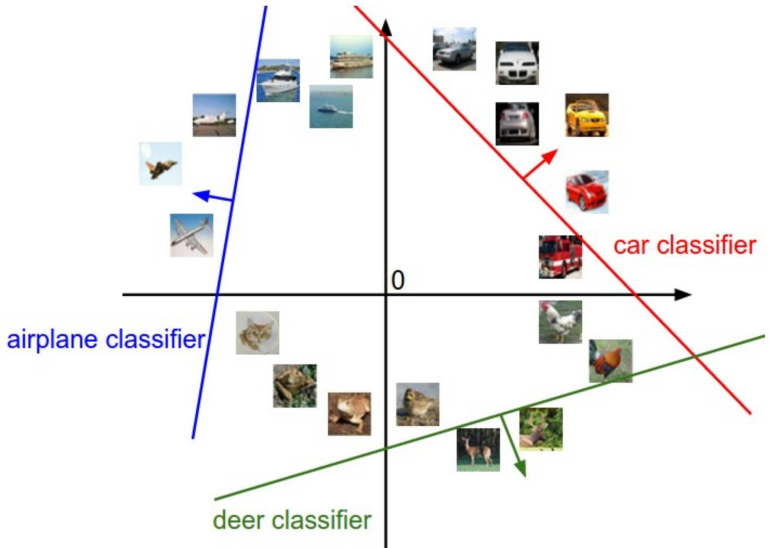
$$f(x) = \sum_{i=1}^{M} g_i(x^{(i)}, y^{(i)})$$

  where $(\forall i \in \{1, \ldots, M\})$ $-g_i \colon \mathbb{R}^P \times \mathbb{R}^Q \to]-\infty, +\infty]$ utility function at node i and $y^{(i)} \in \mathbb{R}^Q$: observations available at node i.

Example: Linear classification

- **Applications:**
  Sentiment analysis from text features
  Handwritten digits recognition
  Gene expression data classification
  Object recognition in images

- **Goal:** Learn linear functions $\ell_k(z) = x_{k,0} + \mathbf{x}_k^\top z$ for dividing the input space into a collection of K regions.



Example: Linear classification

Logistic regression For every $k = 1, \ldots, K-1$,

$$\Pr(G = k | Z = z) = \frac{\exp(x_{k,0} + \mathbf{x}_k^\top z)}{1 + \sum_{\ell=1}^{K-1} \exp(x_{\ell,0} + \mathbf{x}_\ell^\top z)}$$

and

$$\Pr(G = K | Z = z) = \frac{1}{1 + \sum_{\ell=1}^{K-1} \exp(x_{\ell,0} + \mathbf{x}_\ell^\top z)}$$

Binary case: $(\forall i = 1, \ldots, n) \quad y_i = -1$ if $g_i = 1$, and $y_i = +1$ if $g_i = 2$.

$$L(\mathbf{x}) = \sum_{i=1}^{n} \log(1 + \exp(-y_i \mathbf{x}^\top \mathbf{z}_i))$$

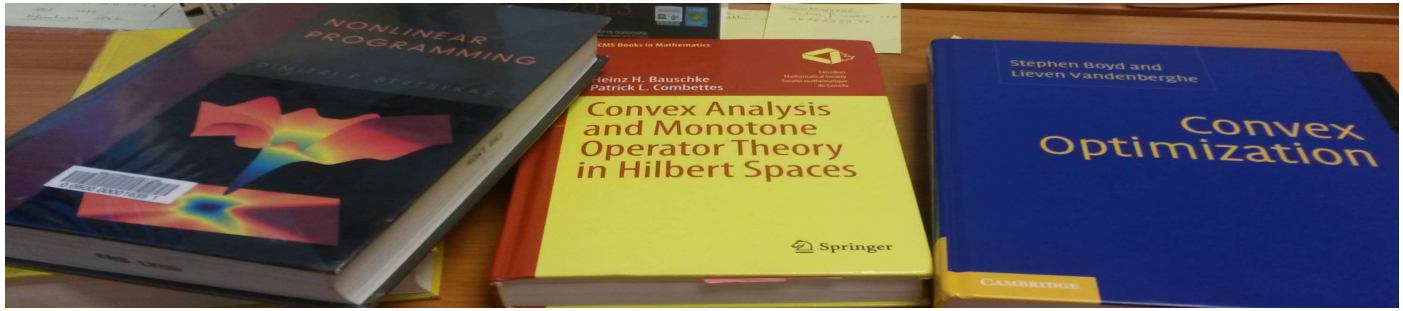
For large datasets (i.e. large n) $\rightsquigarrow$ Need for regularization to avoid over-fitting. We minimize:

$$f(\mathbf{x}) = L(\mathbf{x}) + \lambda \|\mathbf{x}\|_1.$$

Outline

- Introduction to optimization problems
- Background on convex analysis
- Parallel proximal algorithms
- Majorization-Minimization approaches
- Distributed methods

Reference books



- **D. Bertsekas**, Nonlinear programming, Athena Scientic, Belmont, Massachussets, 1995.
- **Y. Nesterov**, Introductory Lectures on Convex Optimization: A Basic Course, Springer, 2004.
- **S. Boyd and L. Vandenberghe**, Convex optimization, Cambridge University Press, 2004.
- **H. H. Bauschke and P. L. Combettes**, Convex Analysis and Monotone Operator Theory in Hilbert Spaces, Springer, New York, 2011.